# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — MATHEMATICS UNIT TEST SET C

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Board Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures; use graph paper where a graph is asked.

## SECTION — A (Objective Type Questions) 12 × 1 = 12

*Choose the correct option and write it in your answer book.*

**1.** The volume of a cone of radius 3 cm and height 7 cm is: 1

**(a)** 66 cm³ **(b)** 22 cm³ **(c)** 198 cm³ **(d)** 132 cm³

**2.** A closed cylinder (radius 7 cm, height 10 cm) has a hemisphere of the same radius on top. Student A says its total surface area is 902 cm²; Student B says 1056 cm². Then: 1

**(a)** Student A is correct **(b)** Student B is correct

**(c)** Both are correct **(d)** Both are wrong

**3.** A student states: "When two solids are joined, the total surface area of the new solid is the sum of their total surface areas." The correct response is: 1

**(a)** Correct

**(b)** Incorrect — the surfaces in contact are not counted

**(c)** Correct only for cubes

**(d)** Cannot be determined

**4.** A cylinder and a cone have equal radii and equal heights. The ratio of their volumes is: 1

**(a)** 2 : 1 **(b)** 1 : 3 **(c)** 1 : 1 **(d)** 3 : 1

**5.** If the mode is 24 and the mean is 60, then the median is: 1

**(a)** 48 **(b)** 42 **(c)** 52 **(d)** 36

**6.** The class mark of a class is 25 and its width is 10. The class is: 1

**(a)** 20–25 **(b)** 15–25 **(c)** 25–35 **(d)** 20–30

**7.** The mode of grouped data is given by: 1

**(a)** $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$ **(b)** $l + \frac{f_1 - f_2}{2f_1 - f_0 - f_2} \times h$ **(c)** $l + \frac{f_1 + f_0}{2f_1 + f_0 + f_2} \times h$ **(d)** $l + \frac{f_1 - f_0}{f_0 + f_2} \times h$

**8.** The mean of the first 10 natural numbers is: 1

**(a)** 6 **(b)** 5.5 **(c)** 5 **(d)** 10

**9.** The probability of an impossible event is: 1

**(a)** 1 **(b)** $\frac{1}{2}$ **(c)** 0 **(d)** −1

**10.** A die is thrown once. The probability of getting a number less than 7 is: 1

**(a)** 1 **(b)** 0 **(c)** $\frac{5}{6}$ **(d)** $\frac{1}{6}$

**11.** A letter is chosen at random from the word ASSASSINATION. The probability that it is a vowel is: 1

**(a)** $\frac{4}{13}$ **(b)** $\frac{7}{13}$ **(c)** $\frac{5}{13}$ **(d)** $\frac{6}{13}$

**12.** A card is drawn from a well-shuffled deck of 52 cards. The probability that it is an ace is: 1

**(a)** $\frac{1}{52}$ **(b)** $\frac{1}{13}$ **(c)** $\frac{4}{13}$ **(d)** $\frac{1}{4}$

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 20–30 words each.*

**13.** Find the volume of a solid formed by a cylinder of radius 2 cm and height 10 cm with a cone of height 3 cm (same radius) on top. **2**

**14.** Find the mode of the data: **2**

| Runs | 0–10 | 10–20 | 20–30 | 30–40 | 40–50 | 50–60 |
|---|---|---|---|---|---|---|
| Batsmen | 5 | 8 | 7 | 12 | 28 | 20 |

**15.** If the mean of the following data is 15.5, find $p$: **2**

| $x$ | 5 | 10 | 15 | 20 | 25 |
|---|---|---|---|---|---|
| $f$ | 3 | 5 | $p$ | 6 | 4 |

**16.** A student says: "The probability of a head in a coin toss is $\frac{1}{2}$, so in 10 tosses exactly 5 heads will come." Justify or refute. **2**

**17.** A box contains cards numbered 1 to 20. A card is drawn at random. Find the probability that the number is **2**
(i) prime
(ii) a multiple of 5.

## SECTION — C (Short Answer Type) — 6 × 3 = 18

*Answer in about 40–60 words each.*

**18.** A hemisphere of radius 4.2 cm is scooped out from one end of a solid cylinder of radius 4.2 cm and height 10 cm. Find the total surface area of the remaining solid. **3**

**19.** Find the median from the cumulative frequency table: **3**

| Marks less than | 10 | 20 | 30 | 40 | 50 | 60 |
|---|---|---|---|---|---|---|
| Students | 5 | 12 | 25 | 38 | 45 | 50 |

**20.** Find the mode of: **3**

| Class | 5–10 | 10–15 | 15–20 | 20–25 | 25–30 |
|---|---|---|---|---|---|
| Frequency | 6 | 11 | 20 | 14 | 9 |

**21.** A lot of 20 bulbs contains 4 defective ones. One bulb is drawn at random. **3**
(i) Find the probability that it is defective.
(ii) If the bulb drawn is not defective and is not replaced, find the probability that the next bulb drawn is defective.

**22.** A child has a die whose six faces show the letters A, B, C, D, E, A. The die is thrown once. Find the probability of getting **3**
(i) A
(ii) D
(iii) a vowel.

**OR**

A bag contains white, black and red balls, 30 in all. The probability of drawing a white ball is $\frac{1}{5}$ and of a black ball is $\frac{1}{2}$. Find the number of red balls. **3**

**23.** A juice glass is cylindrical with inner diameter 5 cm and height 10 cm, but its bottom has a raised hemispherical portion of the same diameter. Find the apparent and actual capacity of the glass. (Use $\pi = 3.14$) **3**

**OR**

A hemispherical depression is cut out from one face of a cubical block of edge $l$ such that the diameter of the hemisphere equals $l$. Find the surface area of the remaining solid. **3**

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**24.** Find the median and mode of the wheat yield of 100 farms: **5**

| Yield (qtl/ha) | 50–55 | 55–60 | 60–65 | 65–70 | 70–75 | 75–80 |
|---|---|---|---|---|---|---|
| Farms | 2 | 8 | 12 | 24 | 38 | 16 |

**OR**

Find the mean, median and mode of the following data and comment on the result. **5**

| Class | 0–10 | 10–20 | 20–30 | 30–40 | 40–50 |
|---|---|---|---|---|---|
| Frequency | 5 | 10 | 20 | 10 | 5 |

**25.** All the black face cards are removed from a pack of 52 cards. A card is drawn at random from the remaining cards. Find the probability of getting **5**

(i) a face card

(ii) a red card

(iii) a black card

(iv) a king.

**OR**

A bag contains 12 balls of which $x$ are white. **5**

(i) If one ball is drawn at random, what is the probability that it is white?

(ii) If 6 more white balls are put in the bag, the probability of drawing a white ball becomes double the earlier probability. Find $x$.

(iii) Find the probability of not drawing a white ball after the 6 balls are added.

## END OF QUESTION PAPER — Total: 50 Marks

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